

Open-system dynamics on a single continuous quantum circuit with a classical cost independent of the number of time steps

Simon Mader¹

¹*Preprint draft — affiliation to be completed*
(Dated: August 24, 2026)

Dilation-based quantum simulation of open quantum systems has a structural weakness that is rarely stated plainly: in essentially every published scheme the classical computer must first produce a substantial part of the answer — a sequence of dynamical maps, a memory kernel, or a window of past states — before the quantum circuit can run at all, and the amount of that classical work grows with the length of the simulation. We remove this dependence. For Markovian evolution a Stinespring dilation turns the channel into one fixed unitary followed by an environment reset; nothing is measured, nothing is re-prepared, and the circuit is deterministic. For non-Markovian evolution we observe that the hierarchical equations of motion (HEOM) already form a closed linear system, so that the exact propagator on the auxiliary-density-operator (ADO) space needs no memory kernel and no memory length K . The single obstruction is that this propagator has spectral radius exactly one but operator norm 32.4 for the four-site FMO model, which destroys any step-wise Sz.-Nagy dilation because post-selection probabilities multiply. We show that this is a property of the inner product rather than of the physics, and remove it in two independent ways: a Lyapunov metric in which the generator is dissipative, and the physical rescaling of the ADOs, which alone reduces the operator norm from 7.7×10^5 to 1.02 at strong coupling. A Krylov compression is then contractive by construction, and the whole evolution runs as one gate repeated on eight qubits with post-selection probability one. Measured against QuTiP’s HEOM solver, the circuit reproduces the four-site FMO dynamics to 4×10^{-7} over 1 ps and 4.7×10^{-6} over 10 ps, with zero classically propagated time steps. We also state carefully what the method does *not* deliver: no quantum advantage, a gate that is far beyond present hardware, and a read-out whose cost is not accounted for.

I. INTRODUCTION

Simulating an open quantum system on a quantum computer means implementing a non-unitary map with unitary hardware. The standard tool is dilation: enlarge the Hilbert space until the map becomes the restriction of a unitary. Both classical constructions are old — Stinespring’s for completely positive maps [1] and Sz.-Nagy’s for contractions [2] — and both have been used repeatedly to put open-system dynamics onto circuits.

There is, however, a recurring and under-advertised catch. A dilation can only be built from a matrix that is already known. In practice this means the classical computer must first generate the object to be dilated, and in the non-Markovian case that object is usually a *sequence*: the dynamical maps $\mathcal{E}_1, \dots, \mathcal{E}_K$, the transfer tensors derived from them [3], or a window of past reduced states. The length K of that sequence controls the accuracy, so improving the result means computing more of the trajectory classically — exactly the work the quantum computer was supposed to do. For the four-site Fenna–Matthews–Olson (FMO) model studied below, reaching an accuracy of 9×10^{-4} over 100 steps by the transfer-tensor route requires 40 of those 100 steps to be produced classically first.

This paper removes that dependence. The central claim is narrow and we state it precisely, because a looser version of it would be wrong:

The classical work is $O(1)$ in the number of time steps n and in the total simulated time

T. A fixed, n -independent set-up produces one unitary, and the circuit then runs for arbitrarily many steps by repeating it.

We emphasise what this does not say. It does not say the classical work is small — it is $O(\mathcal{D}^3)$ in the dimension $\mathcal{D}$ of the extended space, and for the model treated here it is in fact slower than solving the equations of motion outright. It does not claim a quantum advantage. What it does say is that the classical and quantum parts have been cleanly separated: the classical side pays once, and the number of time steps is thereafter carried entirely by the circuit.

Section II treats the Markovian case, where the construction is deterministic and the post-selection problem does not arise at all. Section III treats the non-Markovian case, where it does, and constitutes the main body of the paper. Section IV reports measurements against QuTiP’s HEOM solver [4, 5], and Section VI states the limitations.

II. MARKOVIAN EVOLUTION: ONE GATE AND A RESET

Let Φ be a completely positive trace-preserving (CPTP) map on a d -dimensional system with Kraus operators $\{M_k\}_{k=1}^K$, so that $\sum_k M_k^\dagger M_k = \mathbb{I}$. Introduce an environment $\mathcal{H}_E \cong \mathbb{C}^K$ with orthonormal basis $\{|k\rangle\}$ and

define

$$V : \mathcal{H} \rightarrow \mathcal{H}_E \otimes \mathcal{H}, \quad V|\psi\rangle = \sum_{k=1}^K |k\rangle_E \otimes M_k|\psi\rangle. \quad (1)$$

Proposition 1. *V is an isometry, and $\text{Tr}_E[V\rho V^\dagger] = \Phi(\rho)$ for every state ρ .*

Proof. For the first claim, $\langle\varphi|V^\dagger V|\psi\rangle = \sum_{k,k'} \langle k|k'\rangle \langle\varphi|M_k^\dagger M_{k'}|\psi\rangle = \langle\varphi|(\sum_k M_k^\dagger M_k)|\psi\rangle = \langle\varphi|\psi\rangle$, using trace preservation in the last step. For the second, expanding $V\rho V^\dagger = \sum_{k,k'} |k\rangle_E \langle k'|_E \otimes M_k \rho M_{k'}^\dagger$ and tracing over E gives $\sum_j \sum_{k,k'} \langle j|k\rangle \langle k'|j\rangle M_k \rho M_{k'}^\dagger = \sum_k M_k \rho M_k^\dagger = \Phi(\rho)$. $\square$

Because the columns of $V \in \mathbb{C}^{N \times d}$ ($N = Kd$) are orthonormal, they can be completed by $N - d$ further orthonormal columns to a unitary $U \in \text{U}(N)$ whose first d columns are those of V . Since $|0\rangle_E \otimes |\psi\rangle$ is supported only on those first d coordinates, $U(|0\rangle_E \otimes |\psi\rangle) = V|\psi\rangle$, and therefore

$$\rho_n = \Phi(\rho_{n-1}) = \text{Tr}_E[U(|0\rangle_E \langle 0|_E \otimes \rho_{n-1})U^\dagger]. \quad (2)$$

Equation (2) is the entire Markovian algorithm. One fixed gate U acts on environment plus system; the partial trace is realised on hardware by resetting the environment register. Nothing is measured, nothing is post-selected, nothing is re-prepared between steps, and iterating n times gives $\Phi^n(\rho_0)$ exactly. The success probability is 1 by construction.

For the four-site FMO model with a Lindblad generator this yields 2 system qubits, 4 environment qubits and 16 Kraus operators; the circuit is 100 applications of one gate interleaved with resets. The single classical input is the Kraus set of the *one-step* channel $\Phi(\Delta t)$, computed once.

The reason this is so easy is precisely the reason it does not generalise: a fixed CPTP map iterated on the system alone is a semigroup, hence Markovian. Non-Markovian evolution therefore requires either a coherent memory register or non-trace-preserving operations, i.e. post-selection. We now take the first route, and the post-selection appears as the price of the dilation rather than of the physics.

III. NON-MARKOVIAN EVOLUTION

A. HEOM is already a closed linear system

In the hierarchical equations of motion [6, 7] the reduced density matrix is propagated together with a set of auxiliary density operators (ADOs) that carry the bath correlations. Stacking them into one vector

$$\vec{\Sigma}(\tau) = (\text{vec } \rho_S, \text{vec } \rho_{\mathbf{n}_1}, \dots) \in \mathbb{C}^{\mathcal{D}}, \quad \mathcal{D} = N_{\text{ADO}} d^2, \quad (3)$$

the evolution is local in time, $\partial_\tau \vec{\Sigma} = \mathcal{L} \vec{\Sigma}$, with a time-independent $\mathcal{L}$. With $P = e^{\mathcal{L}\Delta\tau}$ and the projector $\Pi = [\mathbb{K}_{d^2} \ 0 \ \dots \ 0]$ onto the zeroth ADO,

$$\text{vec } \rho_S(k\Delta\tau) = \Pi P^k \vec{\Sigma}(0) \quad \text{exactly, for every } k. \quad (4)$$

Proposition 2. *On the extended space the evolution is a semigroup; all non-Markovianity of ρ_S arises from Π not being invertible. In particular no memory length K appears.*

Proof. $\mathcal{L}$ is time-independent, so $e^{\mathcal{L}(\tau+\sigma)} = e^{\mathcal{L}\tau} e^{\mathcal{L}\sigma}$ and $\vec{\Sigma}_k = P \vec{\Sigma}_{k-1}$ depends on its immediate predecessor only. The reduced maps $\mathcal{E}_k = \Pi P^k \Pi^\top$ fail to compose because $\Pi^\top \Pi = \text{diag}(\mathbb{K}_{d^2}, 0, \dots, 0) \neq \mathbb{K}$ — projecting onto the system block *deletes* the bath memory that the ADOs were carrying. $\square$

This is the point at which the transfer-tensor construction gives away more than it needs to. It starts one level below Eq. (4), from the reduced maps $\mathcal{E}_k$, inverts the convolution to obtain transfer tensors T_m , and truncates the resulting sum at K terms. Both of its costs — the discarded correlations beyond $K\Delta t$ and the K classically propagated states needed to fill the history window — are consequences of having projected in the first place. Keeping the ADOs costs a larger vector and removes K entirely.

B. The obstruction: a bounded trajectory with a large operator norm

Dilating P directly fails, and it is worth being precise about why. For a matrix M with $\|M\|_2 \leq s$, the Sz.-Nagy dilation

$$U = \begin{pmatrix} M/s & \sqrt{\mathbb{K} - MM^\dagger/s^2} \\ \sqrt{\mathbb{K} - M^\dagger M/s^2} & -(M/s)^\dagger \end{pmatrix} \quad (5)$$

is unitary, and projecting one ancilla onto $|0\rangle$ after each of n applications succeeds with probability

$$p_n = \frac{\|M^n x\|^2}{s^{2n} \|x\|^2}. \quad (6)$$

The denominator is a worst case over the whole space applied n times, while the numerator follows the one trajectory that is actually taken. For the four-site FMO HEOM generator at $\Delta t = 10$ fs we measure $\rho(P) = 1.000000$ but $\|P\|_2 = 32.39$, so $s^{2n} = 32.39^{200} \approx 10^{301}$.

Refining the time step does not help.

Proposition 3. *Let $T = n\Delta\tau$ be fixed and $s(\Delta\tau) = \|e^{\mathcal{L}\Delta\tau}\|_2$. Then $n \ln s(\Delta\tau) \rightarrow T \omega(\mathcal{L})$ as $\Delta\tau \rightarrow 0$, where $\omega(\mathcal{L}) = \lambda_{\max}(\frac{1}{2}(\mathcal{L} + \mathcal{L}^\dagger))$ is the numerical abscissa.*

Proof. With $x(\tau) = e^{\mathcal{L}\tau} x_0$, $\frac{d}{d\tau} \|x\|^2 = 2x^\dagger \frac{1}{2}(\mathcal{L} + \mathcal{L}^\dagger)x \leq 2\omega \|x\|^2$, with equality for the top eigenvector of the Hermitian part. Hence $s(\Delta\tau) = 1 + \omega\Delta\tau + O(\Delta\tau^2)$ and $n \ln s = \frac{T}{\Delta\tau}(\omega\Delta\tau + O(\Delta\tau^2)) \rightarrow T\omega$. $\square$

We measure $\omega(\mathcal{L}) = +2.61 \times 10^3 \text{ cm}^{-1}$ and $T = 0.1884 \text{ cm}$, giving $T\omega \approx 4.9 \times 10^2$ and $p \approx e^{-984}$ irrespective of the discretisation. The gap between $\rho(P) = 1$ and $\|P\|_2 = 32$ is the familiar signature of a strongly non-normal operator [8].

It is instructive that the trajectory itself is not bounded either, but in the opposite direction: along the physical path $\|P^n \vec{\Sigma}_0\|/\|\vec{\Sigma}_0\|$ rises to a maximum of 3165 and sits at 1101 after 100 steps. Both factors in Eq. (6) are therefore absurd — 8.2×10^{-303} and 1.2×10^6 — and they very nearly, but not nearly enough, cancel. The reason is that only 5.2×10^{-4} of $\|\vec{\Sigma}_{100}\|$ belongs to the system block; the rest lives in the ADOs, which are bookkeeping objects rather than density matrices and are under no obligation to stay bounded. The Euclidean norm is simply the wrong yardstick on this space.

C. A metric in which the generator is dissipative

Theorem 1 (Lyapunov gauge). *Let $\delta > 0$ be such that $\mathcal{L} - \delta\mathcal{K}$ is Hurwitz. Then*

$$(\mathcal{L} - \delta\mathcal{K})^\dagger W + W(\mathcal{L} - \delta\mathcal{K}) = -\mathcal{K} \quad (7)$$

has a unique solution, it is Hermitian positive definite, and $\|e^{\mathcal{L}\tau}\|_W \leq e^{\delta\tau}$ for all $\tau \geq 0$, where $\|x\|_W = \|W^{1/2}x\|_2$.

Proof. Set $W = \int_0^\infty (e^{(\mathcal{L}-\delta)\tau})^\dagger e^{(\mathcal{L}-\delta)\tau} d\tau$. The integral converges because $\mathcal{L} - \delta$ is Hurwitz, and $x^\dagger W x = \int_0^\infty \|e^{(\mathcal{L}-\delta)\tau} x\|_2^2 d\tau > 0$ gives positive definiteness. Differentiating the integrand and integrating from 0 to ∞ yields Eq. (7), the boundary term at infinity vanishing and the one at zero giving $-\mathcal{K}$. Uniqueness holds because the Sylvester operator $X \mapsto (\mathcal{L} - \delta)^\dagger X + X(\mathcal{L} - \delta)$ has eigenvalues $\bar{\lambda}_i + \lambda_j$ with strictly negative real part. Finally, with $V(\tau) = x(\tau)^\dagger W x(\tau)$ and $\dot{x} = \mathcal{L}x$,

$$V' = x^\dagger (\mathcal{L}^\dagger W + W \mathcal{L}) x = x^\dagger (-\mathcal{K} + 2\delta W) x \leq 2\delta V,$$

and Grönwall gives $V(\tau) \leq e^{2\delta\tau} V(0)$. $\square$

The shift is needed only because $\mathcal{L}$ has the stationary state as an exact zero eigenvalue, which would make the integral diverge along that one direction.

Corollary 1. $\tilde{P} = W^{1/2} P W^{-1/2}$ satisfies $\|\tilde{P}\|_2 \leq e^{\delta\Delta\tau}$ and is similar to P . The gauge therefore introduces no approximation whatsoever: $\tilde{P}^k = W^{1/2} P^k W^{-1/2}$ and $\text{vec } \rho_S(k\Delta\tau) = \Pi W^{-1/2} \tilde{P}^k \tilde{\Sigma}_0$.

Read with $s = e^{\delta\Delta\tau}$, Eq. (6) becomes $p_n = e^{-2\delta T} \|\tilde{P}^n \tilde{\Sigma}_0\|^2 / \|\tilde{\Sigma}_0\|^2$, which is $O(1)$. Measured: $\|P\|_2 = 32.39 \rightarrow \|\tilde{P}\|_2 = 1.000038$ and p_{100} from 9.9×10^{-297} to 0.365, a ratio of 3.7×10^{295} (Fig. 1).

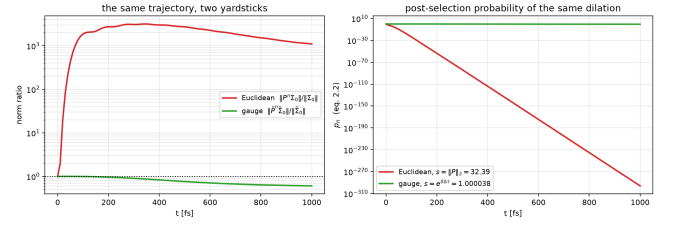


FIG. 1. The same physical trajectory measured with two yardsticks. Left: the Euclidean norm of the ADO vector grows by three orders of magnitude, because almost all of it belongs to the auxiliary operators rather than to ρ_S ; in the gauge it is a contraction. Right: the post-selection probability of the identical Sz.-Nagy dilation, 10^{-297} against 0.365.

TABLE I. Effect of the ADO rescaling (8) on the operator norm of the one-step propagator, four-site FMO, $\Delta t = 10$ fs, hierarchy depth 3.

λ (cm^{-1})	$\ P\ _2$ raw	$\ P\ _2$ rescaled	spread $s_{\max}/s_{\min}$
3	32.39	1.001	1.08×10^5
35	3.12×10^4	1.017	4.18×10^6
100	7.71×10^5	1.023	2.07×10^7

D. Attacking the cause: rescaling the ADOs

The gauge treats the symptom. The cause is that the ADOs, in the conventional HEOM normalisation, differ enormously in magnitude: the operator with multi-index $\mathbf{n}$ carries a prefactor $\prod_k c_k^{n_k}$, and the bath coefficients c_k span orders of magnitude. A unit step in a deep ADO means something numerically very different from a unit step in ρ_S , and the transient growth of $\|P\|_2$ is precisely the propagator moving weight from small-scale to large-scale coordinates.

The standard remedy [9] is the diagonal rescaling

$$\tilde{\rho}_{\mathbf{n}} = \frac{\rho_{\mathbf{n}}}{\sqrt{\prod_k n_k! |c_k|^{n_k}}} \iff \tilde{\mathcal{L}} = D^{-1} \mathcal{L} D, \quad (8)$$

a similarity transformation and hence exact. The zeroth ADO carries the factor 1, so neither the initial state nor the read-out is affected. Its effect is striking (Table I): the operator norm falls from 7.7×10^5 to 1.023 at $\lambda = 100 \text{ cm}^{-1}$. The “norm catastrophe” is thus largely an artefact of the coordinate convention. We note that QuTiP 5 does not apply this rescaling.

Rescaling alone is not sufficient — $1.023^{200} \approx 10^2$ would still be fatal, and $\rho(P) = 1$ still forces $\delta > 0$ — but it leaves the gauge only a residual correction to make, with a condition number many orders of magnitude smaller. A blind numerical alternative is not a substitute: balancing the generator with `scipy.linalg.matrix_balance` reaches only $\|P\|_2 = 2.54$ at $\lambda = 3$ and achieves nothing at all at $\lambda = 100$.

E. Krylov compression becomes free

The gauged propagator still acts on $\mathbb{C}^{2640}$. Compress it onto the Krylov space $\mathcal{K}_m(\tilde{P}, \tilde{\Sigma}_0)$ with an orthonormal Arnoldi basis Q_m [10], $\tilde{P}Q_m = Q_m H_m + h_{m+1,m} q_{m+1} e_m^\top$, $H_m = Q_m^\dagger \tilde{P} Q_m$.

Proposition 4. $\|H_m\|_2 \leq \|\tilde{P}\|_2$ for every m .

Proof. Q_m is an isometry, so $\|H_m y\| = \|Q_m^\dagger \tilde{P} Q_m y\| \leq \|\tilde{P} Q_m y\| \leq \|\tilde{P}\| \|y\|$. $\square$

This is why the gauge must come first. Compressing the raw P gives $\|H_m\| \approx 32$ and a trajectory that diverges to 10^{65} ; compressing $\tilde{P}$ gives a contraction for *every* m , so m ceases to be a trade-off and becomes a pure convergence parameter, costing $\lceil \log_2 m \rceil$ qubits and nothing else. Moreover $H_m^k Q_m^\dagger \tilde{\Sigma}_0 = Q_m^\dagger \tilde{P}^k \tilde{\Sigma}_0$ for $k \leq m-1$, so the first m steps are exact.

F. The circuit

H_m is padded to $m_p = 2^{\lceil \log_2 m \rceil}$ and dilated once by Eq. (5) into a single unitary on $\lceil \log_2 m \rceil + 1$ qubits. No state preparation is required: by construction of the Arnoldi basis the initial vector is $\|\tilde{\Sigma}_0\| e_1$, so the register starts in $|0 \dots 0\rangle$. Each step applies U , measures the ancilla into its own classical bit and resets it; the run is accepted if the record is all zeros.

Proposition 5. *Conditioned on the record $0 \dots 0$, the register state after t steps is $H_m^t y_0 / \|H_m^t y_0\|$, independent of any randomness in the measurements.*

Proof. By induction. If the register is in the pure state $|y_{t-1}\rangle|0\rangle_E$, Eq. (5) gives $U|y_{t-1}\rangle|0\rangle = (M_s|y_{t-1}\rangle)|0\rangle + (D|y_{t-1}\rangle)|1\rangle$ with $M_s = H_m/s$. Obtaining 0 projects onto the first summand, and renormalisation cancels $1/s$, leaving $H_m|y_{t-1}\rangle/\|H_m y_{t-1}\|$, a function of y_{t-1} alone. The reset restores the hypothesis. $\square$

The outcome record is random; the state conditioned on it is not. One accepted shot therefore already carries the exact state, and statistics are needed only for the scalar p_t , which supplies the norm through $\|y_t\| = s^t \sqrt{p_t} \|y_0\|$. Alternatively, trace preservation $\text{Tr} \rho_S(t) = 1$ fixes the same scalar without any statistics. This also dictates how the circuit must be simulated: Qiskit Aer samples mid-circuit measurements in the density-matrix method exactly as in the statevector method, so reading the post-selected branch out of a saved density matrix is Monte-Carlo, whereas `save_statevector(conditional=True)` returns the accepted branch itself.

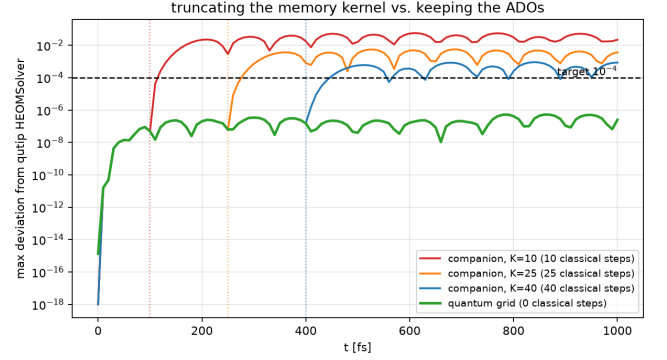


FIG. 2. Deviation from the exact solver over 100 steps. The transfer-tensor route (coloured) improves only by moving more of the trajectory onto the classical computer; the vertical dotted lines mark where each history window ends. The present method (green) uses none.

IV. RESULTS

All numbers are for the four-site FMO Hamiltonian with one Drude-Lorentz Padé bath per site, $\gamma = 50 \text{ cm}^{-1}$, $T = 300 \text{ K}$, $\Delta t = 10 \text{ fs}$, hierarchy depth 3 (165 ADOs, $\mathcal{D} = 2640$), and are measured against QuTiP's HEOMSolver [4] and nothing else.

a. Against the transfer-tensor route. At $\lambda = 3 \text{ cm}^{-1}$ over 100 steps, the companion operator gives 5.66×10^{-2} , 5.58×10^{-3} and 9.03×10^{-4} for $K = 10, 25$ and 40 , requiring 10, 25 and 40 classically propagated steps respectively. Iterating the ADO propagator gives 5.27×10^{-7} with none (Fig. 2). That floor is the reference solver's own integration tolerance: propagating the full 2640×2640 matrix with neither gauge nor compression reproduces the identical figure.

b. Convergence in m . With $\delta = 0.02$: $m = 16$ gives 1.93×10^{-1} (5 qubits), $m = 32$ gives 6.22×10^{-2} (6), and $m \geq 64$ gives 5.27×10^{-7} (7 qubits), with $\|H_m\|_2 = 1.000038$ independent of m as required. Against the exact gauged trajectory the compression error at $m = 128$ is 1.9×10^{-14} over 100 steps and 1.65×10^{-8} over 500.

c. Long runs. Since $\|H_m\|_2 \leq e^{\delta \Delta \tau}$, the iteration cannot blow up. Over 100, 500 and 1000 steps the circuit reproduces the solver to 4.99×10^{-7} , 1.85×10^{-6} and 4.65×10^{-6} , with the post-selection probability settling into a plateau rather than decaying (Fig. 3). The error grows sub-linearly and saturates; the density matrix stays trace-preserving to 2.4×10^{-9} and positive semidefinite to -2×10^{-15} over 5 ps. The circuit itself adds nothing: it reproduces the classical iteration of H_m to 7.7×10^{-10} .

d. The shift δ . The naive expectation is that δ costs only $e^{-2\delta T}$ and is otherwise irrelevant. That is incomplete, because W itself depends on δ : the smaller δ , the more weight the metric places on the stationary state, towards which the trajectory relaxes, so the state loses almost no length in that metric. Measured p_{100} : 0.1995

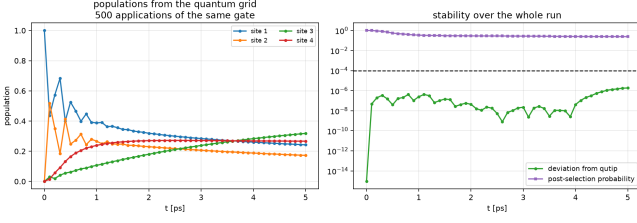


FIG. 3. Populations from the circuit over 500 applications of the same gate (left) and the deviation from the exact solver together with the post-selection probability (right).

TABLE II. Coupling sweep, 100 steps, $\delta = 10^{-5}$, $m = 128$. “Guarantee” is the check $\|\tilde{P}\|_2 \leq e^{\delta\Delta\tau}$. At $\lambda = 1000$ the generator itself is unstable (spectral abscissa $+3.70$).

λ	resc.	cond(W)	vs. solver	p_{100}
3	no	2.87×10^{13}	5.27×10^{-7}	0.9987
3	yes	6.78×10^8	5.27×10^{-7}	1.0000
100	no	4.09×10^{22}	5.80×10^{-8}	0
100	yes	1.60×10^9	5.79×10^{-8}	1.0000
300	no	W not pos. def.		
300	yes	6.79×10^9	2.00×10^{-8}	1.0000
1000	yes	gen. unstable		

at $\delta = 10^{-1}$, 0.8896 at 10^{-3} , and 1.0000 at 10^{-7} , while $e^{-2\delta T}$ moves only from 0.963 to 1.000. Below that, $\text{cond}(W)$ grows as $1/\delta$; at $\delta = 10^{-11}$ it reaches 2.1×10^{19} , the guarantee $\|\tilde{P}\|_2 \leq e^{\delta\Delta\tau}$ is violated (3.56 measured), and p collapses to zero. The practical rule is to reduce δ until $p \approx 1$ and no further; comparing the measured norm against the guaranteed bound is a free validity check.

e. Coupling strength. Table II shows the range. Without rescaling the method fails between $\lambda = 3$ and $\lambda = 100$. With it, everything up to $\lambda = 300 \text{ cm}^{-1}$ runs with $\|\tilde{P}\|_2 = 1.000000$ and $p = 1.0000$, and $\text{cond}(W)$ grows only linearly in λ rather than quadratically. At $\lambda = 1000$ the calculation fails even with rescaling — but for a structural reason: the spectral abscissa of the depth-3 generator is $+3.70 \text{ cm}^{-1}$, so the truncated hierarchy is genuinely unstable, no positive definite W exists, and the reference solver is equally meaningless there. The cure is a deeper hierarchy, not better linear algebra. For comparison, the physically relevant FMO value is $\lambda \approx 35$.

V. WHERE THE WORK HAPPENS

Table III is the point of the paper. Every classical step is performed once and none of them scales with n or T .

The comparison that matters is with the alternatives. The transfer-tensor route pays K propagation steps before the circuit starts, and K must grow with n to hold the accuracy; a direct classical solve pays all n . Here the classical side pays a fixed $O(\mathcal{D}^3)$ and the same $m = 128$

TABLE III. Cost of the non-Markovian construction. $\mathcal{D} = N_{\text{ADO}}d^2$ is the ADO-space dimension, m the Krylov dimension, n the number of steps.

step	cost	grows with n ?
assemble $\mathcal{L}$	sparse	no
Lyapunov solve (7)	$O(\mathcal{D}^3)$	no
$P = e^{\mathcal{L}\Delta\tau}$	$O(\mathcal{D}^3)$	no
Arnoldi	m matvecs	no
dilation (5)	$O(m^3)$	no
propagation	$n \times$ one gate	on the circuit

carries 100, 500 and 1000 steps without modification.

One qualification is required, and it is the one a careful reader will raise first. The Krylov space $\mathcal{K}_m = \text{span}\{\tilde{\Sigma}_0, \tilde{P}\tilde{\Sigma}_0, \dots, \tilde{P}^{m-1}\tilde{\Sigma}_0\}$ contains the first m time steps, and building it costs m matrix–vector products. For $n \lesssim m$ the method therefore saves nothing at all. The claim that survives is the scaling one: m does not depend on n , whereas K does.

VI. WHAT THIS DOES NOT DO

We list the limitations plainly, because several of them are severe.

No quantum advantage. The set-up takes 122 s against the reference solver’s 0.27 s for the same answer. The circuit performs the cheapest part of the problem — repeatedly applying a small, already-known matrix. The result demonstrates that exact non-Markovian dynamics fits on a continuous circuit, not that doing so is currently worthwhile.

The gate is not hardware-executable. U is a dense 256×256 matrix without structure; a Quantum Shannon decomposition [11] needs $\sim 3 \times 10^4$ CNOTs per application and $\sim 3 \times 10^6$ for 100 steps, far beyond present coherence times. We measured 94, 423 and 1783 CNOTs at 4, 5 and 6 qubits, consistent with the $\frac{23}{48}4^n$ scaling.

The read-out is not priced in. $\text{vec } \rho_S = Ry_t$ is a linear functional of the register state, not an observable, and on hardware would require tomography of the Krylov register or a Hadamard test per component. This gap is shared with the transfer-tensor route.

Accuracy is measured against the same truncation. Our 5×10^{-7} is the distance to QuTiP at depth 3 and $N_k = 1$, not to the physics. Depth 4 against depth 3 differs by 2.3×10^{-3} (whereas $N_k = 2$ against $N_k = 1$ differs by only 1.2×10^{-5}), so the hierarchy truncation — shared with the reference — dominates the real error budget by four orders of magnitude.

Memory. The Lyapunov solve needs several dense $\mathcal{D} \times \mathcal{D}$ matrices. Seven sites at depth 2 ($\mathcal{D} = 5880$) is feasible; at depth 3 ($\mathcal{D} = 33\,320$, ~ 18 GB per matrix) it is not. Replacing the dense solve by a low-rank method is the obvious next step, with the caveat that low-rank solvers

return a singular W , which is not directly usable as a metric.

State dependence. The gauge W is universal and is reused, but the Krylov basis is not: a different initial state requires a fresh Arnoldi run (no new Lyapunov solve and no new exponential). The grid is therefore not a universal propagator in the sense of a dynamical map.

VII. CONCLUSION

For Markovian evolution, a Stinespring dilation gives a deterministic circuit: one fixed gate, an environment reset, no measurement and unit success probability. For non-Markovian evolution, the auxiliary density operators already carry the full memory, so the exact propagator requires no memory kernel and no memory length; the only obstruction is that this propagator has operator norm 32 while its spectral radius is 1, and that obstruction lives in the inner product rather than in the physics. A Lyapunov metric removes it exactly, the physical ADO rescaling removes most of it at its source, and a Krylov

compression is then contractive by construction. What remains is one gate on eight qubits, repeated, with post-selection probability one and agreement with the exact solver at the level of that solver's own tolerance.

The classical cost of the construction is independent of the number of time steps. That is a structural statement about where the work sits, not a performance claim: the constant is large, and the resulting gate is far beyond current hardware. The natural continuations are a sparse or low-rank Lyapunov solver, which would lift the memory ceiling, and a block-encoding of $\mathcal{L}$ that never forms P or W densely — in which case the contraction gauge would remain necessary, since the norm problem it solves does not depend on how the gate is built.

ACKNOWLEDGMENTS

This work builds on the Kraus-operator route to non-Markovian dynamics of Ref. 12. Numerical reference dynamics were computed with QuTiP [4, 5]; circuits were built and simulated with Qiskit [13].

-
- [1] W. F. Stinespring, Proc. Am. Math. Soc. **6**, 211 (1955).
 - [2] B. Sz.-Nagy, Acta Sci. Math. (Szeged) **15**, 87 (1953).
 - [3] J. Cerrillo and J. Cao, Phys. Rev. Lett. **112**, 110401 (2014).
 - [4] N. Lambert, T. Raheja, S. Cross, P. Menczel, S. Ahmed, A. Pitchford, D. Burgarth, and F. Nori, Phys. Rev. Research **5**, 013181 (2023).
 - [5] J. R. Johansson, P. D. Nation, and F. Nori, Comput. Phys. Commun. **184**, 1234 (2013).
 - [6] Y. Tanimura and R. Kubo, J. Phys. Soc. Jpn. **58**, 101 (1989).
 - [7] Y. Tanimura, J. Chem. Phys. **153**, 020901 (2020).
 - [8] L. N. Trefethen and M. Embree, *Spectra and Pseudospectra: The Behavior of Nonnormal Matrices and Operators* (Princeton University Press, 2005).
 - [9] Q. Shi, L. Chen, G. Nan, R.-X. Xu, and Y. Yan, J. Chem. Phys. **130**, 084105 (2009).
 - [10] Y. Saad, SIAM J. Numer. Anal. **29**, 209 (1992).
 - [11] V. V. Shende, S. S. Bullock, and I. L. Markov, IEEE Trans. Comput.-Aided Des. Integr. Circuits Syst. **25**, 1000 (2006).
 - [12] A. Seneviratne, P. L. Walters, and F. Wang, ACS Omega **9**, 9666 (2024).
 - [13] Qiskit contributors, [Qiskit: An open-source framework for quantum computing](#) (2023).